

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

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Finding Reasoning Paths :
1. **Anatomical snuff box -> anatomical snuff box -> anatomical snuff box -> anatomical snuff box -> anatomical snuff box**
2. **Aerio venous anastomosis -> anatomical snuff box -> anatomical snuff box -> anatomical snuff box**
3. **Aerio venous anastomosis -> anatomic snuff box artery -> anatomical snuff box -> anatomical snuff box -> anatomical snuff box**

Reasoning Process :
1. The **anatomical snuff box** is a region of the forearm defined by the tendons of the extensor pollicis brevis , longus , and abductor pollicis longus . It houses important vascular and nervous structures , including the **lateral cutaneous nerve** of the forearm and the anatomic snuff box artery , which is part of the **anterior branch of the brachial artery** .
2. The **anatomic snuff box artery** is a superficial artery that runs along the lateral aspect of the hand and is located just lateral to the anatomical snuff box . Its superficiality makes it a potential site for bleeding , especially in cases of trauma .
3. In the context of the described injury (penetrating knife wound to the anatomical snuff box), the **anatomic snuff box artery** is most likely damaged . This artery runs near the surface of the skin and is easily injured in this region .
4. The **brachial artery** is the main artery of the upper limb and gives rise to the anatomic snuff box artery , which branches into the palmar superficial arch . The wound described is consistent with damage to the anatomic snuff box artery , as it lies superficially and is prone to injury in this location .
5. The **brachial artery** is not directly responsible for excessive bleeding in this case , as its major branches (e.g. , anatomic snuff box artery) are more likely to be affected in trauma . However , damage to the brachial artery itself could also result in significant bleeding , depending on the extent of the injury .

Conclusion :
The **anatomical snuff box artery** , a branch of the brachial artery , is most likely responsible for the excessive bleeding in this case . Damage to this artery near the anatomical snuff box region is consistent with the described injury and would result in significant bleeding .

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<answer>
Princeps pollicis artery
</answer>